Humna Sadia

Software Engineer

Islamabad, Pakistan | +92 316 1703289 | i230580@isb.nu.edu.pk
github.com/Hhumna | linkedin.com/in/humna-sadia

SUMMARY

Computer Science undergraduate at FAST-NUCES with hands-on experience across software engineering, mobile development, machine learning, and high-performance computing. Built cross-platform applications with React Native, Flutter, and Firebase; trained and deployed machine-learning models in Python; and engineered concurrent C++ systems using POSIX IPC and CUDA. Strong foundation in data structures, algorithms, object-oriented design, and the full software development lifecycle. Seeking opportunities to contribute to engineering teams across product, infrastructure, or research-oriented work.

EDUCATION

Bachelor of Science in Computer Science Sept 2023 – Present
FAST National University of Computer and Emerging Sciences (FAST-NUCES) Islamabad, Pakistan
Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Systems, Software Engineering, Artificial Intelligence, Parallel & Distributed Computing, Computer Networks.

Higher Secondary — Pre-Engineering (FSc) Sept 2020 – 2022
Kaharian Girls Cadet College Kallar Kahar, Pakistan

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, Dart, SQL, HTML, CSS, Assembly (MASM)

Frontend & Web: React, Next.js, Tailwind CSS, Framer Motion, WordPress

Mobile Development: React Native, Flutter, MVVM Architecture, Cross-Platform Design

Backend & Data: Node.js, Firebase (Firestore, Authentication), REST APIs, Streamlit

Machine Learning: XGBoost, Random Forest, K-Means Clustering, Label Propagation, Genetic Algorithms, Search-Based Planning

Parallel & Systems Programming: CUDA, OpenACC, POSIX IPC, Shared Memory, pthreads, Multithreading, Process Synchronization

Tools & Platforms: Git, GitHub, Docker, Figma, Linux, Cisco Packet Tracer, VS Code

Concepts: Software Development Lifecycle, Algorithm Design, Object-Oriented Design, System Design, UML Modeling, Agile / Scrum

EXPERIENCE

Mobile App Developer Intern July 2025 – August 2025
FAST-NUCES Islamabad, Pakistan

- Designed mobile UI prototypes in Figma including navigation flow, screen hierarchy, and a reusable component library to ensure consistency across the application.
- Developed a cross-platform Point-of-Sale application using React Native with MVVM architecture, separating presentation, logic, and data layers for maintainability.
- Integrated Firebase Authentication and Firestore for real-time data synchronization and secure role-based access control.
- Improved usability through optimized UI layouts and intuitive interaction design across phone and tablet form factors.

Graphic Designer (Remote) January 2025 – Present
Bright Beginnings Academy Lahore, Pakistan

- Design visual assets, social media graphics, and marketing collateral supporting ongoing digital campaigns.
- Maintain brand consistency across deliverables while adapting tone and format for different campaign objectives.

Graphic Designer (Remote) June 2024 – September 2024
SS Consultancy Services Haripur, Pakistan

- Produced promotional materials, presentations, and branding assets for client outreach initiatives.
- Translated written briefs into polished visual deliverables on tight turnarounds.

PROJECTS

Cross-Platform Point-of-Sale Application *React Native, Firebase, MVVM*

- Built a full-featured POS system supporting real-time inventory, order management, and multi-role user access.
- Implemented MVVM architecture with clear separation of view, view-model, and data layers, improving testability and reuse.
- Integrated Firebase Firestore for real-time synchronization and Firebase Auth with role-based claims (admin, cashier, manager).

Chrono Rift — Multi-Process Game Engine *C++, POSIX IPC, pthreads, Docker*

- Engineered three isolated processes (Game Arbiter, Human Interface, AI Strategy) communicating exclusively through POSIX shared memory and unnamed semaphores.
- Implemented thread-per-entity concurrency using pthread mutexes, condition variables, and semaphore-based signaling for safe access to shared state.
- Developed a stamina-based turn-resolution system modeling CPU arrival-time scheduling, applying OS scheduling principles to game logic.
- Integrated POSIX signal handling (SIGUSR1, SIGSTOP, SIGCONT, SIGALRM, SIGTERM) for stun mechanics, ultimate abilities, and graceful shutdown.
- Built deadlock detection and resolution logic monitoring circular resource dependencies across entity threads.

AI-Powered Quiz System *Python, XGBoost, Random Forest, Streamlit*

- Trained an answer-verification model with XGBoost using 18 handcrafted features (TF-IDF, Jaccard similarity, lexical overlap), achieving 99.97% classification accuracy on the validation set.
- Developed a distractor and hint-generation model using n-gram extraction, frequency-based substitution, and a Random Forest hint scorer.
- Implemented unsupervised K-Means clustering reaching 97.5% purity and semi-supervised Label Propagation reaching 99.2% accuracy with only 20% labeled training data.
- Built a four-screen Streamlit application: article input, interactive quiz, graduated hint panel, and analytics dashboard.

KLT Feature Tracker — GPU Acceleration *C++, CUDA, OpenACC*

- Profiled the CPU baseline of the Kanade-Lucas-Tomasi feature-tracking algorithm to identify computational bottlenecks.
- Implemented CUDA kernels with shared-memory tiling and memory coalescing, achieving significant speedup over the CPU baseline.
- Reimplemented the solution using OpenACC directives and benchmarked both, demonstrating tradeoffs between explicit and directive-based GPU parallelism.

Multi-Agent Game Playing in Stochastic Environments *Python, Dear PyGui, Expectiminimax*

- Implemented Expectiminimax search with Alpha-Beta pruning, handling chance nodes for combat dice rolls, minefield events, and environmental stochastic factors.
- Built three AI agent tiers (Expert, Intermediate, Novice) with varying search depths, observation ranges, and evaluation heuristics.
- Developed a real-time visualization GUI for grid display, agent state inspection, and performance analytics.

Warehouse Storage Optimization *Python, Genetic Algorithm*

- Designed a Genetic Algorithm to optimize product placement across 8 warehouse zones satisfying 9 categories of constraints.
- Implemented tournament selection, crossover, and adaptive mutation operators with configurable population size and generation limits.
- Engineered a multi-penalty fitness function evaluating weight capacity, fragile protection, hazardous isolation, temperature control, and product compatibility.

SplitTracker — Shared Expense Management *Flutter, Dart*

- Developed a cross-platform mobile application enabling users to track shared expenses and manage spending categories.
- Designed responsive mobile UI ensuring smooth usability across a range of Android and iOS device sizes.

Multi-Robot Path Planning System *Python, BFS / Uniform-Cost Search*

- Built a priority-based multi-agent pathfinding system with checkpoint sequencing, energy constraints, and collision avoidance via time-step reservations.
- Designed a state-space representation combining position, checkpoint index, and time-step into a unified planning model.
- Supported one-way passage enforcement and other terrain constraints typical of factory and warehouse layouts.

Ideon Research Collaboration Platform *Java, JavaScript, UML*

- Followed a full SDLC approach: requirements gathering, UML class / sequence / activity diagrams, then implementation.
- Designed and implemented backend architecture in Java with role-based access control and structured data workflows.

Zoo Management System *Java, JavaScript, Scrum*

- Developed interactive frontend interfaces and a RESTful service layer for CRUD operations on animal records.
- Applied Scrum methodology with sprint planning and iterative feature delivery.

Graph Isomorphism Detector *Python*

- Implemented a graph isomorphism detection algorithm using optimized traversal and permutation techniques.
- Improved runtime on large graph datasets through pruning strategies that reduce the search space.

GameBoy System — Snake, Wordle, Hangman *C++, Object-Oriented Programming*

- Developed three classic arcade games within a shared object-oriented framework, applying inheritance and polymorphism to reduce duplication.
- Decoupled game logic, input handling, and rendering following SOLID principles.

Network Simulation *Cisco Packet Tracer*

- Configured LAN, WAN, and VLAN architectures using routing protocols (OSPF, RIP).
- Implemented DHCP, DNS, and ACL-based security policies.

KurutKart — Jewellery E-Commerce Website *WordPress*

- Built and deployed a responsive storefront with optimized product catalog, category navigation, and purchase flow.
- Customized WordPress themes and plugins to match brand identity and client-specified workflows.

CERTIFICATIONS & ACTIVITIES

- Flutter Bootcamp — Google Developer Groups on Campus (GDGoC), FAST Peshawar (2024 – 2025)
- Unity Game Development Bootcamp — Google Developer Groups on Campus (GDGoC), PIEAS (April 2025)
- Infyma AI Hackathon — Participant (March 2025)
- Graphic Designing Certificate — Graphix Online Academy (2022)

LANGUAGES

English: Professional working proficiency
Urdu: Native